

Yutao Yang

INCOMING DIRECT-ENTRY PH.D. STUDENT · VIDEO UNDERSTANDING & WORLD MODELS

Beijing / Tianjin, China

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Research interests: video understanding and temporal modeling, world models, vision-language-action learning, and hierarchical planning

Profile

Born Apr. 7, 2005; Chinese nationality; originally from Hongtong, Shanxi, China; full member of the Communist Party of China. Incoming direct-entry Ph.D. student jointly trained by Beijing Zhongguancun College and the College of Excellent Engineers at Nankai University. My research interests include video understanding and temporal modeling, world models, vision-language-action (VLA) learning, and task decomposition with hierarchical planning. I am the second author of an accepted ACM Multimedia 2026 paper and the lead developer of LumenSport.

Education

Beijing Zhongguancun College & College of Excellent Engineers, Nankai University

Beijing / Tianjin, China

DIRECT-ENTRY PH.D. STUDENT, ELECTRONIC INFORMATION

Sep. 2026 – Present

- Research interests: video understanding and temporal modeling, world models, vision-language-action learning, and task decomposition with hierarchical planning.
- Joint study in Electronic Information at Beijing Zhongguancun College and the College of Excellent Engineers, Nankai University.

Shanxi University

Taiyuan, China

B.ENG. IN SOFTWARE ENGINEERING

Sep. 2022 – Jun. 2026

- Required-course GPA: 3.74/4.50; weighted average: 89.70/100; major rank: 7/801 (Top 0.9%); graduate-recommendation rank: 4/801 (Top 0.5%).
- Outstanding Undergraduate Graduate of Shanxi University; National Encouragement Scholarship three times; First-Class Academic Scholarship three times.

Publications

[1] Yandong Liu, **Yutao Yang**, Shengjiao Dong, Wenqiang Zhang, Xiang Li, Mengli Yu, Zhao-Min Chen, and Lingfeng Yang. “Speech-Aware Multimodal Video Slicing for Automated Video Trimming.” *Proceedings of the 34th ACM International Conference on Multimedia (ACM MM)*, 2026. **Accepted; second author; CCF-A conference.** The work introduces SeamSlicer, a multimodal atomic-boundary extraction and refinement framework, and AtomicBench, an atomic-level video-trimming benchmark.

Research Experience

LumenSport: AI-Powered Sports Video Analysis and Automated Editing

Nankai University /

Open-source project

LEAD DEVELOPER / PROJECT LEAD

Sep. 2025 – May 2026

- Led an end-to-end system that transforms raw sports footage into edited highlight reels, focusing on football and basketball while supporting nine sports.
- Designed a layered pipeline for sport classification, round/possession segmentation, highlight scoring, tactical event analysis, player detection and tracking, and automatic reel generation.
- Integrated Swin Transformer, SlowFast, ConvNeXtV2, VideoMAE, YOLO, ByteTrack, and large multimodal models; built player-focused football/basketball processing and Alibaba Cloud OSS support.
- Developed an editing orchestrator with eight storytelling strategies: highlights, funny moments, tactical analysis, technical analysis, comparison, knowledge, match replay, and training.
- Built a Python backend and React/Vite/TypeScript interface covering editing, materials, analytics, and tactical views; released the system under Apache-2.0 at github.com/Yutao-Yang/LumenSport.
- The associated football platform won First Prize in the Football Application Innovation Track of the First National Football Artificial Intelligence Innovation Exchange Conference.

SeamSlicer: Speech-Aware Multimodal Video Slicing

SECOND AUTHOR

Nankai University joint
research
Sep. 2025 – May 2026

- Studied automated video trimming at atomic semantic granularity by jointly modeling visual continuity, narration, multi-speaker dialogue, background music, sound effects, on-screen text, and ambient audio.
- Contributed to the research and experiments for multimodal candidate-boundary extraction, visual refinement, and multi-strategy length control.
- Helped construct and evaluate AtomicBench, a multimodal dataset with atomic-level annotations for visual smoothness and semantic coherence.

Siamese Network-Based Surveillance Video Anomaly Detection

PROJECT LEAD

Shanxi University
Jul. 2024 – Jun. 2025

- Established a Siamese-network baseline for abnormal-behavior recognition, then extended the two-branch architecture into a Quadruplet Network to improve feature discrimination and robustness under lighting changes and occlusion.
- Designed grouped/depthwise-separable convolution blocks, multi-scale Concat and skip connections, and a learnable weighting layer that balances pair and triplet losses through backpropagation.

Fire and Smoke Detection with YOLOv8 and Multidimensional Collaborative Attention

PROJECT LEAD

Shanxi University
Mar. 2023 – Sep. 2024

- Designed parallel dilated-convolution feature enhancement, MCA attention, and a coordinate-attention-guided hierarchical feature-selection pyramid for complex fire scenes.
- Achieved 91.8% mAP on a self-built dataset; improved occluded-target recall by approximately 0.5–0.6 percentage points and small-flame accuracy by 1.7 percentage points.

MakeupVision: AI-Based Virtual Makeup Platform

PROJECT LEAD

Shanxi University Innovation
Project
May 2024 – May 2025

- Led the prototype integrating Light-FaceDetector, a 106-point facial-landmark heatmap model, and Makeup-master semantic segmentation for virtual makeup synthesis.
- Designed ROI-to-landmark cross-layer feature transfer and lip/eye masks for improved robustness under low light and back-lighting; project completion was rated Good.

Honors and Awards

NATIONAL-LEVEL COMPETITIONS

Jun. 2026	First Prize , First National Football AI Innovation Exchange Conference, Football Application Innovation Track	China
Jul. 2025	Second Prize , National 3D Digital Innovation Design Competition 17th Anniversary Elite League	China
May 2025	Second Prize , National College Business Elite Challenge, Brand Planning National Finals	China
Jun. 2025	Third Prize , 16th Lanqiao Cup National Finals, Python Programming, University Group B	China
May 2025	Third Prize , 2025 Group Programming Ladder Tournament National Finals	China

PROVINCIAL-LEVEL COMPETITIONS

Jun. 2025	Special Prize , National 3D Digital Innovation Design Competition, Shanxi Division	Shanxi
Aug. 2025	Silver Award , China International College Students' Innovation Competition, Shanxi	Shanxi
May 2025	First Prize , 16th Lanqiao Cup, Python Programming, University Group B	Shanxi
May 2025	First Prize , 2025 Group Programming Ladder Tournament, Shanxi	Shanxi
Jun. 2025	Second Prize , 18th Chinese Collegiate Computing Competition, Digital Media Game and Interaction	Shanxi
Apr. 2024	Second Prize , 15th Lanqiao Cup, Python Programming, University Group B	Shanxi
Jan. 2024	Second Prize , 15th Chinese Mathematics Competition, Non-Mathematics Group A	Shanxi

Jun. 2025	Third Prize , 16th Lanqiao Cup Visual Art Design, Static Poster	<i>Shanxi</i>
May 2025	Third Prize , 2024 National Undergraduate Mathematical Contest in Modeling, Shanxi	<i>Shanxi</i>
Aug. 2024	Third Prize , 17th Chinese Collegiate Computing Competition, Software Applications and Development	<i>Shanxi</i>
Apr. 2023	Third Prize , 14th Lanqiao Cup, C/C++ Programming, University Group B	<i>Shanxi</i>
Jul. 2025	Excellence Award , 27th China Robotics and Artificial Intelligence Competition, Provincial Selection	<i>Shanxi</i>
Dec. 2023	Excellence Award , 2023 National Undergraduate Mathematical Contest in Modeling, Shanxi	<i>Shanxi</i>

UNIVERSITY-LEVEL COMPETITIONS

May 2025	First Prize , 15th Market Survey and Analysis Competition, Shanxi University Selection	<i>Taiyuan</i>
Jul. 2024	First, Second, and multiple Third Prizes , China International College Students' Innovation Competition, Shanxi University	<i>Taiyuan</i>
May 2024	First Prize , 10th New-Form Software Competition, 3D Digital Art Teaching System	<i>Taiyuan</i>
May 2024	Second Prize , 2024 Shanxi University Mathematical Modeling Competition	<i>Taiyuan</i>
May 2024	Second Prize , 10th New-Form Software Competition, Ember Vision	<i>Taiyuan</i>
Apr. 2024	Second Prize , 14th Market Survey and Analysis Competition, Shanxi University Selection	<i>Taiyuan</i>
Sep. 2023	Three Third Prizes , 9th Internet+ Innovation and Entrepreneurship Competition, Shanxi University	<i>Taiyuan</i>
May 2023	Third Prize , 2023 Shanxi University Mathematical Modeling Competition	<i>Taiyuan</i>

Scholarships and Distinctions

Jun. 2026	Outstanding Undergraduate Graduate , Shanxi University	<i>Taiyuan</i>
Oct. 2025	National Encouragement Scholarship , Ministry of Education / Shanxi University	<i>China</i>
Nov. 2024	National Encouragement Scholarship , Ministry of Education / Shanxi University	<i>China</i>
Nov. 2023	National Encouragement Scholarship , Ministry of Education / Shanxi University	<i>China</i>
Oct. 2025	First-Class Academic Scholarship , Shanxi University	<i>Taiyuan</i>
Nov. 2024	First-Class Academic Scholarship , Shanxi University	<i>Taiyuan</i>
Nov. 2023	First-Class Academic Scholarship , Shanxi University	<i>Taiyuan</i>
May 2024	Three-Good Student , Shanxi University	<i>Taiyuan</i>
May 2023	Three-Good Student , Shanxi University	<i>Taiyuan</i>
May 2026	Highest Year-End Performance Tier , Nankai University Research Internship	<i>Tianjin</i>

Certifications, Skills, and Activities

Programming	Python, C/C++, Java, MATLAB, SQL, JavaScript/TypeScript
AI / Vision	PyTorch, OpenCV, NumPy, Pandas, scikit-learn
Engineering	Linux, Git, React, Vite, LaTeX, OSS, Java web deployment, Unity, 3ds Max, SolidWorks, SPSS, Visio
Certification	Huawei HCIA-Kunpeng Application Developer; valid through Apr. 19, 2027
English	CET-4: 427 (Jun. 2023); CET-6: 424 (Dec. 2024)
Data Science	Kaggle Stanford RNA 3D Folding: rank 331/1516 (Top 21.8%)
Technical Writing	CSDN AI creator; more than 400,000 cumulative views

Leadership, Volunteering, and Practical Experience

Calligraphy and Painting Association, Shanxi University

ORGANIZATION AND PUBLICITY OFFICER

Taiyuan

Oct. 2023 – Sep. 2024

- Regularly organized calligraphy and painting exhibitions, member activities, and publicity campaigns, coordinating event preparation and on-site execution.
- Practices Chinese brush calligraphy, specializing in Slender Gold script and regular script.

Class Safety and Mental-Health Service

SAFETY AND MENTAL-HEALTH COMMITTEE MEMBER

Shanxi University

Sep. 2022 – Jun. 2026

- Served as class safety committee member, communicating safety information, conducting routine awareness activities, and coordinating related matters.
- Served as class mental-health committee member, supporting classmates, assisting with well-being activities, and facilitating communication when needed.

In-Person Campus and Community Volunteering During Undergraduate Study

VOLUNTEER

Shanxi University and local community

Sep. 2023 – Jun. 2026

- Completed 202 cumulative hours of in-person volunteer service during undergraduate study.

Independent Teaching and Application Mentoring

IN-PERSON TUTOR AND ONLINE GRADUATE-RECOMMENDATION MENTOR

Taiyuan / Online

Sep. 2023 – Jun. 2026

- Completed more than 300 hours of in-person tutoring for middle-school, high-school, and lower-division university students, helping them strengthen fundamentals and improve academic performance.
- Provided online graduate-recommendation mentoring, helping students plan applications, refine materials, and apply to competitive programs at leading Chinese universities.